

PLambda Cosmology Model Summary Report

This report summarises the recent model fitting results using Pantheon+SH0ES supernovae (N=1701), BAO data, and variations of LCDM, KdS, and FR cosmologies. The focus is on Bayesian Information Criterion (BIC) model comparison and variable interpretation.

1. Results Overview

Key findings from the latest runs:

- FR with supernova magnitude evolution ($\epsilon_M \approx -0.10$) significantly improves fit quality over flat Λ CDM ($\Delta\text{BIC} \approx -40$).
- Models with $\Lambda=0$ (pure KdS or FR no- Λ) are disfavoured due to BAO tension ($\Delta\text{BIC} > +150$).
- Curvature (Ω_k) alone does not provide a better fit than FR with ϵ_M .
- H_0 values cluster around 72–73 km/s/Mpc for FR/ Λ CDM, but drop to ~ 50 –53 for KdS solutions, which the data disfavour when BAO included.

2. Model-by-Model Breakdown

Flat Λ CDM (SN+BAO)

- H_0 : ≈ 73.39
- Ω_m : ≈ 0.263
- Ω_Λ : ≈ 0.737
- Ω_k : 0 (flat)
- ϵ_M : 0 (no evolution)
- BIC: ≈ 2113.55
- Comment: Baseline reference; decent fit but not optimal.

Curved Λ CDM (SN+BAO)

- H_0 : ≈ 73.19
- Ω_m : ≈ 0.224
- Ω_Λ : ≈ 0.667
- Ω_k : $\approx +0.11$
- ϵ_M : 0
- BIC: ≈ 2118.66
- Comment: Slight curvature improves χ^2 but BIC penalises.

FR varM log1pz (flat)

- H_0 : ≈ 72.95
- Ω_m : ≈ 0.252
- Ω_Λ : ≈ 0.748
- ε_M : ≈ -0.10
- BIC: ≈ 2070.73
- Comment: Strongest performer; $\Delta\text{BIC} \approx -43$ vs flat ΛCDM .

FR varM only (flat, with Λ)

- H_0 : ≈ 72.75
- Ω_m : ≈ 0.214
- Ω_Λ : ≈ 0.695
- ε_M : ≈ -0.10
- BIC: ≈ 2069.70
- Comment: Best balance; fits both SN and BAO.

FR varM only ($\Lambda=0$)

- H_0 : ≈ 70.1
- Ω_m : ≈ 0.0
- Ω_Λ : 0
- Ω_k : $\approx +1.0$
- ε_M : ≈ -0.087
- BIC: ≈ 2273.35
- Comment: Disfavoured; BAO χ^2 huge (≈ 230).

KdS flat (SN+BAO)

- H_0 : ≈ 52.75
- Ω_m : ≈ 0.40
- Ω_Λ : ≈ 0.60
- A_{acc} : ≈ 0.91
- n_{acc} : ≈ 0.82
- BIC: ≈ 2116.75
- Comment: High BAO χ^2 ; acceptance fraction low.

3. Variable Glossary

Below is a detailed glossary of all variables and parameters used:

- H_0 : Hubble constant today [km/s/Mpc], sets distance scale.
- Ω_m (Ω_m): Matter density fraction today, evolves as $(1+z)^3$.
- Ω_Λ (Ω_Λ): Cosmological constant density fraction.
- Ω_k (Ω_k): Curvature parameter, $\Omega_k = 1 - \Omega_m - \Omega_\Lambda$.
- χ^2 : Chi-square goodness of fit statistic.
- AIC: Akaike Information Criterion; penalises #params mildly.
- BIC: Bayesian Information Criterion; penalises #params strongly.
- A_{acc} : Amplitude of effective acceleration term in KdS/FR.
- n_{acc} : Index of acceleration term; implies effective w_{eff} .
- γ_c (γ_c): Variable speed of light slope: $c(z) = c_0(1 + \gamma_c z)$.
- ϵ_M (ϵ_M): Supernova absolute magnitude evolution parameter. $\Delta M(z) = \epsilon_M f(z)$.
- μ (μ): Distance modulus; $\mu = 5 \log_{10}(DL/10\text{pc}) + \Delta M(z)$.
- $E(z)$: Dimensionless Hubble parameter.
- D_L : Luminosity distance [Mpc].
- dM-mode: Choice of evolution form: linear_z or log1pz.

4. Acronym Summary

- SN: Supernova
- SNe: Supernovae (plural)
- BAO: Baryon Acoustic Oscillations
- CC: Cosmic Chronometers ($H(z)$ measurements)
- CMB: Cosmic Microwave Background
- FR: Full Relativity model (generalised Kerr–de Sitter with variable-c and variable-M options)
- Λ CDM: Lambda Cold Dark Matter model
- KdS: Kerr–de Sitter model
- AIC: Akaike Information Criterion
- BIC: Bayesian Information Criterion
- PV: Peculiar Velocity
- JWST: James Webb Space Telescope
- SH0ES: Supernovae H_0 for the Equation of State (Riess et al.)

5. Best Run Script

The best performing run (FR varM-only with Λ free, flat) used the following script:

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' --  
model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat'
```

```
--cov '/Users/boyde/Downloads/Pantheon+SH0ES_STAT+SYS.cov' --  
bao '/Users/boyde/Downloads/bao_long.csv' --rd 147.09 --flat --  
fix_A_acc --fix_n_acc --fix_gamma_c --sampler emcee --nwalkers 64 --  
nsteps 4000 --nburn 1000 --nint 240 --tight-priors --corner --seed  
76 --verbose --out 'plamb_runs/FR_varMonly_withLambda_SNBAO'
```
